An independent exact recheck of the $n = 6$ non-transitive dice theorem

(a recheck of a result by others; no novelty is claimed)

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-11-01

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** All six corrections are applied (Section 7). No human audit exists. There is no Lean in this paper.

Abstract

T. Easlick (arXiv:2610.40321, 29 Oct 2026) studies n-sided dice with faces in $\{1, \ldots, n\}$ and the directed graph G_n whose edges are the pairs where one die beats the other with the smallest possible advantage. His Theorem 1 states that for $n = 6$ there are exactly two uniform minimal non-transitive cycles of maximum length 6, and none of length n for $n = 3, 4, 5$. **The theorem is Easlick's; we claim nothing new.** Two independent implementations (a producer's numpy code for $n \leq 9$ and an auditor's C code for $n \leq 10$) reproduce every graph statistic for $n = 3, \ldots, 10$ in exact integer arithmetic; for $n = 6$ both 6-cycles and the absence of 4-, 5- and 7-cycles are reproduced with every edge exactly 19/36; the cycles printed in the paper for $n = 7, 8, 9, 10$ (lengths $14, 29, 10, 110$) verify exactly. Since a cycle longer than n exists for each $n = 7, \ldots, 10$, Conjecture 1 is open only for $n \geq 11$ under the reading of the proof of Theorem 1 (and false at $n = 7$ under the alternative reading). Tier [C] for the exact verification, Tier [X] for searches and context controls.

What this is NOT. Not a new result; not a proof of Conjecture 1 or any statement for $n \geq 11$; not a formal proof (no Lean); not a comparison of the vertex lists of the $n \geq 7$ strongly connected components with the paper's appendices (only sizes and counts); not an exhaustive prior-art search; not human-audited.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code (integers, `Fraction`) with the argument written down, not formalised. [X]: searches and context controls, not exhaustive. Identifiers `DICE-·-nnnn` refer to `ledger/ledger.json`.

1 Definitions

A *die* is a multiset of n faces in $\{1, \ldots, n\}$ (repeats allowed; the same multiset is the same vertex), each face with probability $1/n$; there are $\binom{2n-1}{n}$ dice. Die x *beats* y if $W(x, y) = \#\{(i, j) : x_i > y_j\} > n^2/2$ (strict; ties do not count). The minimal winning count is $W = \lfloor n^2/2 \rfloor + 1$ out of n^2 pairs, i.e. probability $\frac{1}{2} + \frac{1}{n^2}$ for even n and $\frac{1}{2} + \frac{1}{2n^2}$ for odd n (Easlick eq. 1). The graph G_n has an edge $x \to y$ iff $W(x, y) = \lfloor n^2/2 \rfloor + 1$; a *uniform minimal* cycle is a directed cycle in G_n. "Maximum length n" is not formally defined in the paper; we read it, through the proof of Theorem 1 (two 6-cycles exhibited, a 7-cycle excluded in the 7-vertex components), as "an n-cycle exists and no longer cycle exists". The alternative reading is "an n-cycle exists". ([L], `DICE-L-0001`.) This is not the maximin problem of maximising the minimal winning probability.

2 Graph statistics, $n = 3, \ldots, 10$ ([C])

Table 1 lists the statistics. The producer used numpy (count-vector form of the double sum, `scipy` SCCs, brute-force double-sum cross-check for $n \le 6$). The auditor, without running or reading producer code, wrote C using face histograms and cumulative counts ($W(x, y) = \sum_v c_x[v]\, C_y[v-1]$) with an iterative Tarjan SCC; 200,000 random pairs per n agreed with the brute-force sum (no mismatch). All rows equal the paper's numbers (Section 3 and Appendices B–J, including 2115 SCCs at $n = 8$). The $n = 10$ row is the auditor's only.

n	dice	edges	SCCs	nontrivial SCC (vertices, edges)
3	10	3	10	none
4	35	67	35	none
5	126	620	126	none
6	462	5491	450	two: $(7, 10)$ and $(7, 10)$
7	1716	89153	636	one: (1081, 53389)
8	6435	956307	2115	one: (4321, 674108)
9	24310	12528131	3860	one: (20451, 10933011)
10	92378	157244770	10588	one: (81791, 147620052)

Table 1: Graph statistics of G_n, reproduced by two implementations ($n \le 9$) and by the auditor ($n = 10$). No self-loops.

Involution. $f_n(x) = (n + 1 - x_n, \ldots, n + 1 - x_1)$ is an involution on dice and sends every edge $a \to b$ to $f(b) \to f(a)$ (checked on all edges for $n = 3, \ldots, 10$ by the auditor: no violation; by the producer for $n \le 6$). Hence f is an anti-automorphism of G_n. For $n = 3, 4, 5$ every SCC is trivial, so G_n has no directed cycle at all, in particular no n-cycle (`DICE-C-0001`, `-0002`).

3 The case $n = 6$ ([C])

Exhaustive simple-cycle enumeration (producer, and the auditor's own DFS) inside the two 7-vertex SCCs gives, in each, exactly three 3-cycles and one 6-cycle, and none of length 4, 5 or 7; in G_6 altogether six 3-cycles and exactly two 6-cycles. Every edge of each 6-cycle equals exactly $19/36 = \frac{1}{2} + \frac{1}{36}$ (`Fraction`). The map f swaps the two 6-cycles (applied to one and reversed it is a rotation of the other). The 1-based lexicographic vertex labels of the two SCCs, $\{88, 89, 151, 207, 217, 277, 291\}$ and $\{118, 119, 160, 235, 286, 385, 391\}$, equal those of the paper's Appendix E. With $n = 3, 4, 5$ cycle-free, $n = 6$ is the smallest $n \ge 3$ admitting an n-cycle, and its longest cycle has length 6: Theorem 1 is reproduced (`DICE-C-0003`).

4 $n \ge 7$ and Conjecture 1 ([C] and [X])

The cycles printed in the paper for $n = 7, 8, 9, 10$ (lengths $14, 29, 10, 110$) were parsed from its text and checked in print order with exact fractions: all dice are distinct and every edge has the minimal probability (25/49, 33/64, 41/81, 51/100). The $n = 10$ check is the auditor's only. Under the reading of Section 1, one exact-checked cycle longer than n shows that the property "the longest cycle has length n" fails, so it fails at $n = 7, 8, 9$ and also at $n = 10$; the $n = 10$ case needs only the 110-cycle, not the 157-million-edge graph. Easlick's Conjecture 1 ($n = 6$ is the largest n with a uniform minimal cycle of maximum length n) is therefore open only for $n \ge 11$ under this reading. Under the alternative reading "an n-cycle exists" the conjecture is false at $n = 7$: the auditor's randomized walk search in G_7 found exact-checked cycles of *every* length from 3 to 28, including 7, 8 and 14 ([X]: the search is not exhaustive; each cycle found is verified exactly). We do not prove the conjecture for any $n \ge 11$ and do not decide it (`DICE-C-0004`, `DICE-X-0001`).

5 Controls

Positive (exact): Efron's four dice give 2/3 on all four edges; the Moon–Moser triple $(2,4,9),(1,6,8),(3,5,7)$ gives 5/9 on each edge; the paper's 6-cycles have six edges of exactly 19/36. Negative: all 62 single-face ± 1 perturbations of a paper 6-cycle (faces kept in [6]) break the uniform minimal cycle; the edge predicate rejects the near-miss pairs $x = (1,1,2,5,5,5)$, $y = (1,1,1,1,1,5)$ with $W = 20$ and $y = (1,1,1,2,2,5)$ with $W = 18$ (run by the auditor; see Section 7). Context ([X], orthogonal to the claim, producer only): the best 3-cycle winning probability among m-sided dice in $[m]$ is 3/5 ($m = 5$), 7/12 ($m = 6$), 29/49 ($m = 7$), all below $(\sqrt{5}-1)/2$; $m = 4$ has no 3-cycle.

6 Literature note

The closest prior setting we found is Conrey, Gabbard, Grant, Liu and Morrison, *Intransitive Dice* [2]: "proper" n-sided dice with faces in $[n]$ and face sum $n(n+1)/2$ (counted by OEIS A076822); Easlick drops the sum constraint, does not cite it, and Conrey et al. do not study the minimal-advantage graph. Also related, and not examined here beyond the abstract: Angel–Davis (arXiv:1610.08595; uniform sets with advantage $\frac{1}{2} + \frac{1}{2k^2}$, general labels), Hur–Kim (2006.12866), Schaefer–Schweig (1602.00969, 1706.08986), Rooney (2505.21950). None of these treats n-cycles in G_n, and we found no earlier statement of the $n = 6$ theorem, but the search was limited (a few alphaXiv and web queries), so prior work is not excluded ([L], `DICE-L-0002`).

7 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them on the audit's six corrections. (1) The $n = 8$ SCC count 2115 is stated in the paper's Appendix H and matches (the producer wrote "not stated"). (2) The conclusion for $n \geq 7$ is as in Section 4: one exact-checked cycle longer than n settles each of $n = 7, 8, 9, 10$; open only for $n \geq 11$; false at $n = 7$ under the alternative reading. (3) The $n = 10$ graph *was* independently reproduced (by the auditor) and matches Appendix J; the producer had listed it as not done. (4) The preregistered $W = 20$ near-miss control was in fact run on a $W = 36$ pair, and the key `ctrl_predicate_rejects_W20` in `extra_results.json` is mislabelled; the auditor ran $W = 20$ and $W = 18$ near-misses and both are rejected. (5) Provenance: `PREREG_ITEM.md` (sha256 `b3425a39...0051765`) was committed in the same commit as the results, so version control cannot show that it was written first (self-attested; hash recorded); its line "copy in data/" is stale, the paper's text is kept outside the repository, and the corresponding hash in `result.json` refers to an external file. (6) The literature note of Section 6.

8 Limitations and what is not done

- AI audit only; no human audit; no Lean.
- Not verified: $n \geq 11$ and any proof of Conjecture 1; the full vertex lists of the $n \geq 7$ SCCs in the paper's appendices (only sizes and counts); the maximin context controls (not recomputed by the auditor); an exhaustive prior-art search.
- The meaning of "maximum length n" is our reading of the proof, not a definition in the paper.
- Trusted components: Python integers and `fractions`, numpy/scipy (producer), a C compiler (auditor).
- Deviation note: the preregistration is self-attested (Section 7, item 5).

9 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code, summaries and logs are in `data/code`, `data/summaries`, `data/logs` (`.log.txt`); the auditor's C source (`dg.c.txt`), Python script and small logs are in `data/audit`. The paper's full text is third-party and is not

shipped. Inputs are copies of `results/proposal6/2026-11-01/DICE-6/` and of the auditor's scratch directory; nothing was committed, pushed or uploaded.

References

[1] T. Easlick, *Computer-Assisted Results for n-Sided Non-Transitive Dice Taking Face Values in* $[n]$, arXiv:2610.40321 (v1, 29 Oct 2026), unrefereed. ([L])

[2] B. Conrey, J. Gabbard, K. Grant, A. Liu, K. Morrison, *Intransitive Dice*, arXiv:1311.6511 (v1 25 Nov 2013; v2 7 Jul 2016). ([L])